MODULE 4: MATHEMATICAL LOGIC:

Propositional and Predicate Logic,

Propositional Equivalences

Normal Forms,

Predicates and Quantifiers

Nested Quantifiers,

Rules of Inference

MATHEMATICAL LOGIC

- The study of formal systems used to represent statements, determine their truth values, and establish whether arguments are logically valid.
- Logic is important in computer science because it forms the foundation of:
  - Programming and algorithms
  - Database queries
  - Artificial intelligence
  - Digital circuit design
  - Formal verification
  - Automated reasoning
  - Discrete mathematics

BASIC CONCEPTS

- **Proposition:** a declarative statement that is either **true or false**, but not both.

- Examples:
  - "5 is greater than 3." → True
  - "10 is an odd number." → False
  - "Kochi is in Kerala." → True
- The following are **not propositions**:
  - "What is your name?" – Question
  - "Close the door." – Command
  - "x + 5 = 10" – Depends on the value of x

PROPOSITIONAL LOGIC

- A proposition is a collection of declarative statements that has either a truth value "true” or a truth value "false".
- A propositional consists of propositional variables and connectives.
- We denote the propositional variables by capital letters (A, B, C etc).
- The connectives connect the propositional variables.
- Examples:
  - "Man is Mortal", it returns truth value “TRUE” as T. •
  - "12 + 9 = 3 − 2", it returns truth value “FALSE” as F.
- Example not a Proposition
  - "A is less than 2".

LOGICAL CONNECTIVES

- Propositions can be combined using **logical connectives**.

Connective	Symbol	Meaning
Negation	¬P	NOT P
Conjunction	P ∧ Q	P AND Q
Disjunction	P ∨ Q	P OR Q
Exclusive OR	P ⊕ Q	P XOR Q
Implication	P → Q	IF P, THEN Q
Biconditional	P ↔ Q	P IF AND ONLY IF Q

NEGATION (¬)

- – The negation of a proposition A (written as ¬A) is false when A is true and is true when A is false

P	¬P
T	F
F	T

Example:

P: "Today is Monday.“

¬P: "Today is not Monday."

Truth Tables: A truth table shows all possible truth values of a proposition.

For n propositions, the number of combinations is:2^n

CONJUNCTION – AND - ∧

- The AND operation of two propositions A and B (written as A ^ B) is true if both the propositional variable A and B is true.

A	B	A ∧ B
True	True	True
True	False	False
False	True	False
False	False	True

Example:

p: "I attend the class."
q: "I submit the assignment."

p ∧ q :"I attend the class **and** submit the assignment."

DISJUNCTION – OR- V

- The OR operation of two propositions A and B (written as A v B) is true if at least any of the propositional variable A or B is true.

A	B	A ∨ B
T	T	T
T	F	T
F	T	T
F	F	F

Example:

- p: **I will have tea.**
- q: **I will have coffee.**

p v q :**I will have tea OR coffee.**

EXCLUSIVE OR – XOR -⊕

- The exclusive OR is true when **exactly one** of p and q is true.
  - p⊕q

P	Q	P ⊕ Q
T	T	F
T	F	T
F	T	T
F	F	F

Example:

P: The student chooses Python.
Q: The student chooses Java.

Statement:
The student can choose either Python or Java, but not both.

IMPLICATION P->Q

- The implication:P->Q is read as: "If p, then q.“
  - It is false only when p is true and q is false.

P	Q	P → Q	¬P ∨ Q
T	T	T	
T	F	F	
F	T	T	
F	F	T	

Example:
P: "A student studies.“
Q: "The student passes.“
P->Q "If a student studies, then the student passes."

Equivalence: $p \rightarrow q \equiv \neg p \vee q$

BICONDITIONAL - ↔

- The biconditional is:P ↔ Q means: "p if and only if q.“
- It is true when p and q have the same truth value.

P	Q	P ↔ Q	P→Q	Q→P	(P→Q) ∧ (Q→P)
T	T	T			
T	F	F			
F	T	F			
F	F	T			

- **Equivalence:** P↔Q (P Q) (Q P)

WELL FORMED FORMULAS(WFF)

- The well formed formulas(WFFs) or statement formulas or logic formulas are defined recursively (or inductively) as below.
- Propositional variables P,Q,R,... and propositional constants F,T are well formed formulas. They are known as primitive WFFs.
- If P and Q are WFFs then ¬Q, ¬P and Q ∧ P, QVP, P->Q, P<->Q are also WFFs.
- All WFFs are obtained by the above procedures applied a finite number of times.
- For example, the following are WFFs

$$p \wedge (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

$$(p \rightarrow q) \rightarrow (q \rightarrow p)$$

EXERCISE

- Construct the truth table for the formula PV~Q

- Construct the truth table for the formula (PVQ)V~P

EXERCISE

- Using the statements

  R: Mark is Rich

  H: Mark is Happy

  Write the following statements in symbolic form:

  a) Mark is poor but happy
  b) Mark is rich or Unhappy
  c) Mark is neither rich nor happy
  d) Mark is poor or he is both rich and unhappy

TAUTOLOGY, CONTRADICTION AND CONTINGENCY

- A tautology is a formula which is always true for every value of its propositional variables. Eg: $p \vee \neg p$
- Prove $[(A \rightarrow B) \wedge A] \rightarrow B$ is a tautology

TAUTOLOGY, CONTRADICTION AND CONTINGENCY

- A contradiction is a formula which is always false for every value of its propositional variables. Eg: $p \wedge \neg p$
- Prove $(A \vee B) \wedge [(\neg A) \wedge (\neg B)]$ is a contradiction

TAUTOLOGY, CONTRADICTION AND CONTINGENCY

- A proposition that is **sometimes true and sometimes false** is called a contingency.
- Example: p ∧ q
- Prove (A V B) ∧ (¬A) is a contingency

PROPOSITIONAL EQUIVALENCES

- Two propositions P and Q are logically equivalent if they have the same truth value for every possible combination of their variables.
  - P≡Q

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

We say that the statements P and Q are logically equivalent if their truth tables are identical.

LAWS OF PROPOSITIONAL LOGIC:

S.No	Name of Laws	Primal Form	Dual Form
1	Idempotent Law	$p \vee p \equiv p$	$p \wedge p \equiv p$
2	Identity Law	$p \vee F \equiv p$	$p \wedge T \equiv p$
3	Dominant Law	$p \vee T \equiv T$	$p \wedge F \equiv F$
4	Complement Law	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
5	Commutative Law	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
6	Associative Law	$p \vee (q \vee r) \equiv (p \vee q) \vee r$	$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$
7	Distributive Law	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
8	Absorption Law	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
9	De Morgan's Law	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$
10	Double Negation Law	$\neg(\neg p) \equiv p$	-

INVERSE, CONVERSE, AND CONTRA-POSITIVE

- A conditional statement has two parts − **Hypothesis** and **Conclusion**
  - If you do your homework, you will not be punished.
- **Inverse** –An inverse of the conditional statement is the negation of both the hypothesis and the conclusion.
  - If you do not do your homework, you will be punished
- **Converse**−The converse of the conditional statement is computed by interchanging the hypothesis and the conclusion.
  - If you will not be punished, you do not do your homework.
- **Contra-positive** –The contra-positive of the conditional is computed by interchanging the hypothesis and the conclusion of the inverse statement.
  - If you will be punished, you do your homework.

- Show that [~P∩(~Q ∩R)V(Q ∩R)V(P ∩R)]≡R

- Show that $P \leftrightarrow Q \equiv (PVQ) => (P \cap Q)$

- Prove that $P \rightarrow (Q \rightarrow P) \equiv \sim P \rightarrow (P \rightarrow Q)$

- Show that [(PVQ)~(~P∩(~QV~R))]V(~P ∩ ~Q)V(~PV~R)

NORMAL FORMS

- **Elementary Product:** If p and q are any two atomic variables, then

  $p, \neg p \wedge q, \neg q | \wedge p, \neg p \wedge \neg q$ are some examples of elementary products.

- **Elementary Sum:** If P and Q are any two atomic variables, then

  $p, \neg p \vee q, \neg q \vee p, \neg p \vee \neg q$ are some examples of elementary sums.

- A **Normal Form** is a standard or structured way of writing a logical expression using specific combinations of **AND (∧), OR (∨), and NOT (¬)**.
  - A propositional expression is in **Disjunctive Normal Form** if it is an **OR (∨) of one or more AND (∧) terms**.
    - (A ∧ B) ∨ (C ∧ D) - AND terms connected by OR
  - A propositional expression is in **Conjunctive Normal Form** if it is an **AND (∧) of one or more OR (∨) terms**.
    - (A ∨ B) ∧ (C ∨ D) - OR terms connected by AND

- Obtain Disjunctive Normal forms of (i) $P \cap (P \rightarrow Q)$ (ii) $\sim(P \vee Q) \leftrightarrow (P \cap Q)$

- Obtain Conjunctive Normal forms of
  - (i)P→ (Q∨P)
  - (ii) ~(PVQ)↔(P ∩Q)

PREDICATES

- **Predicate Logic**, also called **First-Order Predicate Logic**, extends propositional logic by allowing us to represent **objects, their properties, relationships, and quantities**.
- While propositional logic deals with complete statements such as: "John is a student."
- Predicate logic can represent a general statement such as: "Every student studies.“
  - Propositional logic, P:Ravi is a student but not easily "All students are intelligent.“
- A **predicate** is a statement containing one or more variables whose truth value depends on the value assigned to the variables.
  - Eg: P(x):x>5 has P(x) as the predicate
- A predicate becomes a **proposition** when a specific value is assigned to its variable.

COMPONENTS OF PREDICATE LOGIC

- Variables : Variables represent unspecified objects.
  - Eg: Student (x)
- Constants : A constant represents a particular object or person.
  - Eg: Student (Ravi)
- Predicates: Predicates represent **properties or relationships**.
  - **Unary predicate:** Student(x)
  - **Binary predicate:** Likes(x,y)
  - **Ternary predicate:** Between(x,y,z)
- Logical connectives : ¬, ∧ , ∨ , →, ↔
- Functions
- Quantifiers

FUNCTIONS

- Simple statement function : An expression with a predicate symbol and an individual variable
  - Example: x is a man expressed as H(x); x is a Mortal expressed as M(x)
  - Roy is a man expressed as H(Roy) (Substitution Instance)
- Compound Statement Function: Obtained by combining one or more simple statement functions and logical connectives.
  - Example: x is a Man and x is Mortal expressed as $H(x) \wedge M(x)$
  - X is taller than y expressed as G(x,y)
- Unary, binary & Ternary functions:
  - Unary – Student(x)
  - Binary – Likes(x,y)
  - Ternary – Between(x,y,z)

QUANTIFIERS

- Quantifiers indicate **how many elements** of the domain satisfy a predicate.
  - Universal quantifier - ∀
    - for all
    - for every
    - Eg: $\forall x\, P(x)$ read as P(x) is true for every x in the domain.
    - Every student has an ID
      - $S(x) = x$ is a student , $I(x) = x$ has an ID
      - $\forall x(S(x) \rightarrow I(x))$
  - Existential quantifier - ∃
    - there exists
    - for some
    - Eg: $\exists x\, P(x)$ read as There exists at least one x for which P(x) is true.
    - Some student has scored more than 90.
      - ∃x(S(x)∧ Score(x)>90)
- Every student passed
- Some student passed

NESTED QUANTIFIERS

- When more than one quantifier appears in a statement, we have **nested quantifiers**.
  - $\forall x \exists y\, P(x, y)$ is read as For every x, there exists some y such that P(x,y) is true.
- Order of Quantifiers Matters:
  - Let: P(x,y)=x has sent an email to y
    - $\forall x \exists y\, P(x, y)$ - Every person has sent an email to someone.
    - $\exists y \forall x\, P(x, y)$ - There is one particular person to whom everyone has sent an email.

EXAMPLE

- All students are hardworking. $\forall x(Student(x) \rightarrow Hardworking(x))$
- Some students are hardworking. $\exists x(Student(x) \wedge Hardworking(x))$
- No student is absent. $\forall x(Student(x) \rightarrow \neg Absent(x))$
- Some student is not present. $\exists x(Student(x) \wedge \neg Present(x))$
- Every student who studies and attends class passes.

$$\forall x[(Student(x) \wedge Studies(x) \wedge Attends(x)) \rightarrow Passes(x)]$$

Inference rules in PL

Modens Ponens $\dfrac{\alpha \Rightarrow \beta, \alpha}{\beta}$

And-elimination: from a conjuction any conjunction can be inferred: $\dfrac{\alpha \wedge \beta}{\alpha}$

Example:

Statement-1: "If I am sleepy then I go to bed" ==> P→ Q

Statement-2: "I am sleepy" ==> P

Conclusion: "I go to bed." ==> Q.

Hence, we can say that, if P→ Q is true and P is true then Q will be true.

Proof by Truth table:

P	Q	$P \rightarrow Q$
0	**0**	**0**
0	**1**	**1**
1	**0**	**0**
1	**1**	**1**

Modus Tollens:

The Modus Tollens rule state that if P → Q is true and **¬ Q is true, then ¬ P** will also true.

Notation for Modus Tollens: $\frac{P \rightarrow Q,\ \sim Q}{\sim P}$

Statement-1: "If I am sleepy then I go to bed" ==> P→ Q

Statement-2: "I do not go to the bed."==> ~Q

Statement-3: Which infers that "**I am not sleepy**" => ~P

Proof by Truth table:

P	Q	$\sim P$	$\sim Q$	$P \rightarrow Q$
0	**0**	**1**	**1**	**1**
0	**1**	**1**	**0**	**1**
1	**0**	**0**	**1**	**0**
1	**1**	**0**	**0**	**1**

Hypothetical Syllogism:

The Hypothetical Syllogism rule state that if P→R is true whenever P→Q is true, and Q→R is true.

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

Example:
Statement-1: If you have my home key then you can unlock my home.
P→Q
Statement-2: If you can unlock my home then you can take my money.
Q→R
Conclusion: If you have my home key then you can take my money.
P→R

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$P \rightarrow R$	
0	0	0	1	1	1	←
0	0	1	1	1	1	←
0	1	0	1	0	1	
0	1	1	1	1	1	←
1	0	0	0	1	1	
1	0	1	0	1	1	
1	1	0	1	0	0	
1	1	1	1	1	1	←

Disjunctive Syllogism:

The Disjunctive syllogism rule state that if P v Q is true, and ¬P is true, then Q will be true.

Notation of Disjunctive syllogism: $\frac{P \vee Q,\ \neg P}{Q}$

Example:

Statement-1: Today is Sunday or Monday. ==>PVQ

Statement-2: Today is not Sunday. ==> ¬P

Conclusion: Today is Monday. ==> Q

Proof by truth-table:

P	Q	$\neg P$	$P \vee Q$
0	**0**	**1**	**0**
0	**1**	**1**	**1**
1	**0**	**0**	**1**
1	**1**	**0**	**1**

Simplification:

The simplification rule state that if **P∧ Q** is true, then **Q or P** will also be true.

Notation of Simplification rule: $\frac{P \wedge Q}{Q}$ Or $\frac{P \wedge Q}{P}$

Proof by Truth-Table:

P	Q	$P \wedge Q$
0	**0**	**0**
1	**0**	**0**
0	**1**	**0**
1	**1**	**1**

Resolution:

The Resolution rule state that if P∨Q and ¬ P∧R is true, then Q∨R will also be true.

Notation of Resolution $\frac{P \vee Q, \ \neg P \wedge R}{Q \vee R}$

Proof by Truth-Table:

P	¬ P	Q	R	$P \vee Q$	¬ P∧R	$Q \vee R$	
0	1	0	0	0	0	0	
0	1	0	1	0	0	1	
0	1	1	0	1	1	1	←
0	1	1	1	1	1	1	←
1	0	0	0	1	0	0	
1	0	0	1	1	0	1	
1	0	1	0	1	0	1	
1	0	1	1	1	0	1	←

RESOLUTION

Example1:

Premise 1: The student studies or attends a tutorial.

Premise 2: The student does not study or attends a tutorial.

Conclusion: Therefore, the student attends a tutorial.

RESOLUTION

Example 2:

Given the premises:

1. $p \rightarrow q$
2. $q \rightarrow r$
3. p

Prove that r is true.

RESOLUTION

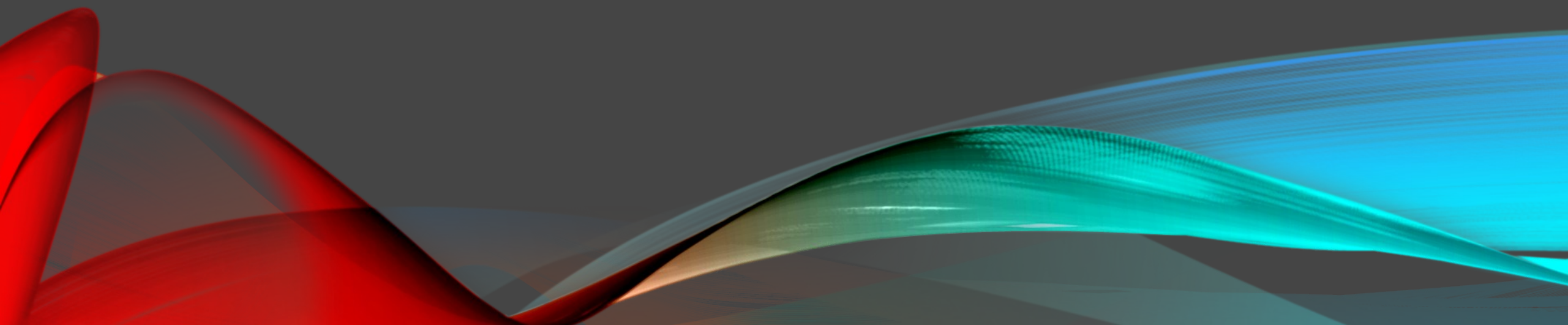
Example 3

Given:

1. $p \vee q$
2. $\neg p$
3. $q \rightarrow r$

Prove r.

If A works hard, then either B or C will enjoy themselves. If B enjoys himself, then A will not work hard. If D enjoy himself, then C will not. Therefore, if A works hard, D will not enjoy himself. Show that these statements constitute a valid argument.



THANK YOU

QP Pattern

Answer Sheet

15 mins before exam

Hall ticket, ID card, Formal wear